EDUBURNOUT AI: EARLY RISK DETECTION USING MACHINE LEARNING

A Project Report

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"The single greatest cause of happiness is gratitude."
-Auliq-Ice

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Abstract

Academic burnout and cognitive overload are increasingly prevalent among students in higher education, driven by mounting academic pressures, information overload, and inadequate coping mechanisms. Early identification of these conditions is critical to enabling timely intervention and supporting student well-being. However, traditional approaches to detection rely on self-reported surveys and retrospective assessments, which are often inaccurate, delayed, and intrusive. There is a growing need for automated, non-intrusive systems capable of predicting burnout and cognitive overload in real time using observable behavioural signals.

This project proposes a machine learning-based framework for the early prediction of academic burnout and cognitive overload by analysing student behavioural patterns. The system collects and processes multi-modal behavioural data including study session durations, interaction logs, assignment submission patterns, browsing behaviour, and response times. Feature engineering techniques are applied to extract meaningful indicators such as irregularity in study schedules, declining engagement trends, and task-switching frequency, which are known correlates of burnout and overload.

Classification models including Random Forest, Support Vector Machines, and LSTM-based deep learning networks are trained and evaluated on labelled student behavioural datasets. The system outputs early warning predictions at multiple risk levels, enabling educators and counsellors to intervene proactively. Experimental results demonstrate that the proposed approach achieves high prediction accuracy while maintaining low false-positive rates. The framework is designed to be lightweight, privacy-preserving, and deployable within existing Learning Management Systems, making it a practical tool for student wellness support in educational institutions.

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Chapter 1

Introduction

Academic burnout and cognitive overload are among the most pressing challenges facing students in higher education today. The rapid expansion of academic curricula, increasing competitive pressures, digital information saturation, and the blurring of boundaries between study and personal life have collectively contributed to a growing mental health crisis among university students. Burnout, characterised by emotional exhaustion, depersonalisation, and a reduced sense of personal accomplishment, has been shown to significantly impair academic performance, mental well-being, and long-term career outcomes [1].

Cognitive overload, a closely related phenomenon, occurs when the volume and complexity of information presented to a student exceeds the processing capacity of working memory. When students are consistently exposed to tasks that demand more cognitive resources than they possess, their ability to learn, retain, and apply knowledge deteriorates significantly [2]. Left undetected, both burnout and cognitive overload can escalate into serious psychological conditions including anxiety disorders and depression.

The proliferation of digital learning environments, Learning Management Systems (LMS), and wearable devices has created an unprecedented opportunity to capture rich, continuous, and objective behavioural data from students. Patterns such as study session frequency and duration, assignment submission timings, login frequency, scrolling behaviour, response latencies, and interaction logs provide a real-time window into a student's cognitive and emotional state. When analysed using machine learning techniques, these behavioural signals can serve as powerful early indicators of academic burnout and cognitive overload [4].

This project proposes a comprehensive framework for the early prediction of academic burnout and cognitive overload through the automated analysis of student behavioural data. The system collects multi-modal behavioural features, applies feature engineering and selection techniques, and trains multiple machine learning classifiers to generate early warning predictions. The

framework is designed to integrate seamlessly with existing LMS platforms and to operate in a privacy-preserving manner, making it practical for real-world deployment.

It is important to recognise that burnout and cognitive overload, while conceptually distinct, are deeply interconnected in practice. Cognitive overload is fundamentally a short-term, task-level phenomenon rooted in the information-processing limits of working memory, whereas burnout is a longer-term, affective and motivational state that develops over weeks or months of sustained strain. In practice, however, the two conditions frequently reinforce one another in a feedback loop: repeated episodes of cognitive overload leave a student feeling perpetually behind, which erodes motivation and self-efficacy, which in turn makes subsequent learning tasks feel even more cognitively demanding than they objectively are. A prediction system that focuses on behavioural signals is well positioned to capture this feedback loop, because many of the same observable indicators, such as declining engagement, erratic study timing, and rising task-switching frequency, are shared precursors of both conditions.

The feasibility of behaviour-based early warning systems has also been shaped by a broader shift in how higher education is delivered. The accelerated adoption of hybrid and fully online learning models over the past several years has meant that a much larger share of a student's academic activity now leaves a digital trace than was the case a decade ago. Lecture attendance, resource access, formative quizzes, discussion forum participation, and assignment submission increasingly occur through LMS platforms rather than in person. This shift is double-edged: it has removed many of the informal, face-to-face cues that instructors historically relied upon to notice a struggling student, such as visible fatigue or withdrawal in the classroom, while simultaneously making available a much richer stream of passive behavioural data than existed before. The framework proposed in this project is designed specifically to take advantage of this shift, substituting the informal observational cues that are now harder to obtain with structured, continuously collected behavioural signals.

1.1 Drawbacks of the Existing Solution

Traditional approaches to identifying at-risk students rely heavily on self-reported surveys, periodic academic assessments, and one-on-one counselling sessions. These methods suffer from significant limitations:

1. **Reactive Rather than Proactive:** Existing methods act only after a problem has become visible. By the time a student is formally identified as experiencing burnout, the condition may already be severe and difficult to reverse [3].

2. **Reporting Bias and Lack of Granularity:** Self-reported data is prone to bias and recall errors, and these methods lack the granularity needed for early and accurate detection.
3. **Infrequent, Blanket Survey Instruments:** Consider a student who experiences a sudden, sharp decline in engagement following an unexpected personal setback, such as a family emergency or a health issue, midway through a semester. If the institution's only mechanism for detecting distress is a single end-of-semester wellness survey, this student's difficulties may go entirely unnoticed until final grades are already affected.
4. **Impractical at Scale:** Standardised psychometric instruments such as the Maslach Burnout Inventory remain the gold standard for clinical assessment, but their use in large-scale monitoring is impractical due to the time and effort required for administration, the risk of survey fatigue, and the inability to provide real-time assessments [22].
5. **Costly and Intrusive Physiological Monitoring:** Physiological approaches are promising but costly and intrusive, and they require specialised hardware that is not yet widely available in educational settings [23].

The points outline the key challenges and limitations of current approaches to detecting academic burnout and cognitive overload.

The following diagram shows the overview of the proposed system in order to overcome the above-mentioned drawbacks of the existing detection approaches.

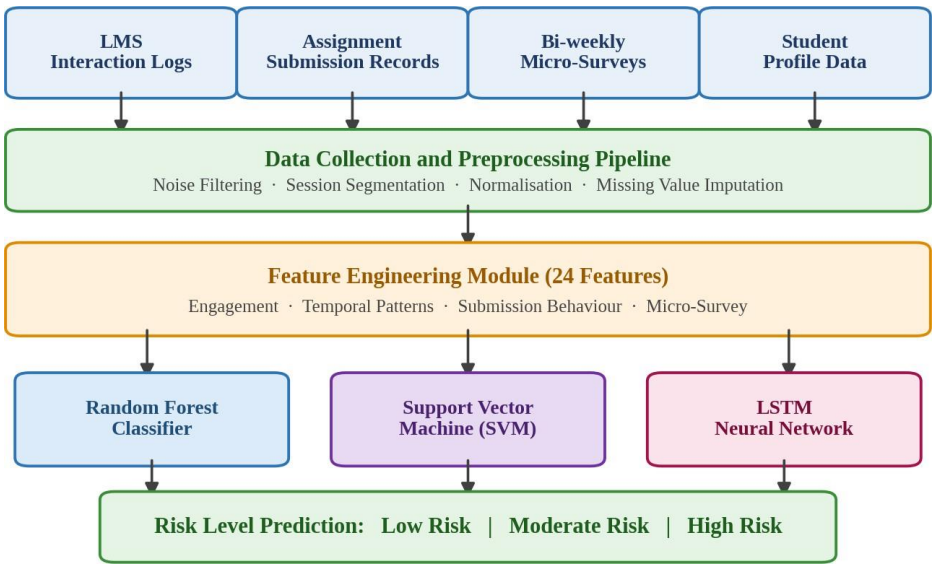


Figure 1.1: Overall System Architecture for Early Burnout and Cognitive Overload Prediction

1.2 Motivation

Several studies have reported that over 40% of university students experience clinically significant levels of burnout during their academic careers [5]. The consequences extend beyond academic performance, affecting physical health, interpersonal relationships, and post-graduation well-being. Despite this, most institutions lack automated, scalable systems capable of identifying students at risk before their condition worsens.

The availability of digital behavioural data offers a unique opportunity to address this gap. Unlike self-reports, behavioural data is objective, continuous, and collected passively without interrupting the student. Machine learning models trained on such data can learn complex non-linear patterns that are invisible to human observers, making them well-suited for early prediction tasks in educational settings.

Beyond the direct impact on individual students, institutions themselves carry a strong motivation to address burnout at scale. Elevated burnout within a cohort is associated with lower retention, higher withdrawal and repeat-enrolment rates, and reduced satisfaction scores in student surveys, all of which have downstream effects on an institution's academic standing, funding, and accreditation outcomes. From a purely operational perspective, even a modest improvement in early identification of at-risk students can translate into meaningful reductions in attrition, making investment in automated monitoring systems financially as well as ethically justifiable for university administrations.

A further motivation stems from the limitations of infrequent, blanket survey instruments. A continuously operating behavioural monitoring system, in contrast to a single end-of-semester wellness survey, is capable of flagging a sudden change in activity pattern within days, creating a window for timely advisor outreach that a periodic survey simply cannot provide.

1.3 Scope

The scope of this project is deliberately bounded so that its goals remain well-defined and achievable within the available data and timeframe.

1.3.1 Target Population and Data Sources

This project focuses on undergraduate students enrolled in technology and science programmes, where academic pressures and information density are particularly high. The behavioural data considered includes LMS interaction logs, assignment submission patterns, and self-reported stress indicators collected through periodic micro-surveys embedded in the learning platform.

The scope is limited to the prediction stage; the design of intervention strategies following a positive prediction is considered future work.

1.3.2 Exclusions

Several related problems are explicitly excluded from the scope of this project. The framework does not attempt to predict burnout in primary or secondary school students, whose developmental stage, academic structures, and available digital trace data differ substantially from the undergraduate context studied here. Graduate research students, whose work patterns are typically project-based and loosely scheduled rather than organised around a fixed LMS-driven curriculum, are also outside the current scope, as the behavioural signals used by the system assume a reasonably structured course and assignment schedule. Likewise, learning environments that operate largely outside a Learning Management System, such as purely in-person programmes with minimal digital infrastructure, are not addressed, since the system's feature extraction pipeline depends fundamentally on the availability of LMS interaction logs.

1.3.3 Rationale for the Chosen Focus

The decision to focus specifically on undergraduate technology and science programmes was driven by two practical considerations. First, these programmes tend to have comparatively dense curricula, frequent formative assessment, and heavy reliance on digital submission platforms, which together produce a richer and more regular stream of behavioural data than is typically available in less digitally intensive disciplines. Second, prior literature reviewed in Chapter 2 indicates that burnout prevalence is particularly high in technically demanding fields, making this population both data-rich and practically significant for early intervention.

1.4 Aim and Objectives

Building on the motivation outlined above, this project is guided by a set of concrete technical and practical objectives. These objectives collectively define what the system must achieve in order to be considered a viable early-warning tool, and they directly shape the design and evaluation decisions presented in the later chapters of this report. The primary objectives of this project are as follows:

1. **Feature Identification:** To identify and extract meaningful behavioural features from student activity logs that are indicative of academic burnout and cognitive overload.

2. **Model Development and Comparison:** To develop and compare multiple machine learning models including Random Forest, Support Vector Machine (SVM), and Long Short-Term Memory (LSTM) networks for burnout prediction.
3. **Performance Evaluation:** To evaluate the performance of the proposed models using standard classification metrics including accuracy, precision, recall, F1-score, and AUC-ROC.
4. **Lightweight, Modular Design:** To design a lightweight, modular prediction framework that can be integrated into existing Learning Management Systems.
5. **Real-World Validation:** To validate the framework on a real-world student behavioural dataset and demonstrate its effectiveness for early warning generation.

Each of these objectives maps onto a specific chapter later in this report: feature identification and extraction is addressed in Chapter 3, model development and comparison in Chapter 3, quantitative evaluation in Chapter 4, and the practicalities of LMS integration and real-world validation are discussed throughout Chapters 3, 4 and 5. Framing the objectives this way ensures that every design decision made in the system can be traced back to a stated project goal, rather than being an arbitrary implementation choice.

1.5 Definition of Key Terms

To ensure clarity and consistency throughout this report, the following key terms are defined:

- **Academic Burnout:** A state of chronic physical and emotional exhaustion resulting from prolonged exposure to demanding academic workloads, characterised by reduced motivation, feelings of cynicism toward one's studies, and a perceived decline in academic competence.
- **Cognitive Overload:** A condition that arises when the total cognitive load imposed by a learning task or environment exceeds the available capacity of the student's working memory, resulting in degraded task performance and reduced information retention.
- **Behavioural Analysis:** The systematic examination of observable student actions and patterns such as login frequency, submission timing, and resource access, to infer underlying cognitive or emotional states.
- **Learning Management System (LMS):** A software platform used by educational institutions to deliver, manage, and track educational content, assignments, and student interactions. Examples include Moodle, Canvas, and Blackboard.

- **Feature Vector:** A numerical representation of a student's behavioural state for a given time window, constructed by aggregating and transforming raw interaction events into a fixed-length array of features suitable for input to a machine learning model.
- **Risk Level:** A categorical label assigned by the prediction system to a student for a given time window, indicating the estimated likelihood of experiencing burnout or cognitive overload. Three levels are used: Low Risk, Moderate Risk, and High Risk.
- **Class Imbalance:** A characteristic of a labelled dataset in which the number of examples belonging to one class, such as High Risk, is substantially smaller than the number of examples belonging to other classes, which can bias a naively trained classifier toward the majority class.
- **Explainability (XAI):** The property of a machine learning system that allows its predictions to be interpreted and justified in terms understandable to a human user, typically through techniques such as SHAP values or attention weight visualisation, rather than being treated as an opaque black box.

1.6 Significance of the Study

The significance of this study lies at the intersection of educational technology, mental health support, and artificial intelligence.

1.6.1 Educational Significance

From an educational perspective, early identification of at-risk students enables institutions to allocate counselling resources proactively rather than reactively, potentially preventing academic failure and withdrawal. Studies have shown that students who receive timely support during early burnout stages are significantly more likely to recover full academic engagement within a single semester compared to those who receive support only after severe deterioration [3].

1.6.2 Technological Significance

From a technological perspective, this project demonstrates the feasibility of deploying lightweight machine learning models within existing institutional infrastructure without requiring additional hardware or significant computational resources. The modular design of the proposed system means that individual components such as the feature extractor or the prediction model can be upgraded independently as the underlying technology evolves.

1.6.3 Societal Significance

From a societal perspective, the project contributes to the broader goal of making higher education more equitable and humane. Students from disadvantaged backgrounds, first-generation university students, and students with limited access to informal support networks are disproportionately affected by burnout and have less access to support services. An automated early warning system embedded in the LMS they already use daily provides a safety net that is available equally to all students regardless of their social circumstances.

1.6.4 Ethical Significance

There is also an important ethical significance to approaching this problem through passive behavioural monitoring rather than more invasive alternatives. Physiological monitoring approaches, discussed in more detail in Chapter 2, require wearable hardware and continuous biometric data collection, which raises substantial privacy and consent concerns and is typically unaffordable at institutional scale. By contrast, a system built on data that institutions already collect for administrative purposes, combined with a lightweight, opt-in micro-survey, offers a markedly less intrusive path to the same early-warning capability, provided it is deployed with the governance safeguards discussed in Section 2.1.11 and Chapter 5.

1.6.5 Economic Scalability

Finally, the study carries a significance tied to economic scalability. Once trained, the marginal cost of generating a daily risk prediction for an additional student is extremely small, as demonstrated by the computational performance figures presented in Chapter 4. This stands in sharp contrast to counsellor-led or survey-administered screening, whose costs scale roughly linearly with the number of students assessed. A system with a low and largely fixed marginal cost per additional student is far more attractive for institutions seeking to extend wellness support to their entire student body rather than a small subset who proactively seek help.

1.7 Organisation of the Report

The remainder of this report is organised as follows. Chapter 2 presents a review of literature on academic burnout, cognitive overload, and machine learning approaches in educational data mining, together with existing detection systems, ethical considerations, and the research gap addressed by this project. Chapter 3 formally states the problem and describes the project flow and methodology, including the dataset, data pipeline, feature engineering, prediction models, training procedure, validation strategy, and system architecture. Chapter 4 describes the tools and technologies used and presents the experimental results and analysis. Chapter 5 concludes

the report, discusses limitations, outlines directions for future work, and offers recommendations for institutional deployment.

Chapter 2

Literature Survey

This chapter reviews the existing literature on academic burnout, cognitive overload, educational data mining, and machine learning-based prediction systems. The review is organised into thematic areas: theoretical foundations of burnout and cognitive overload, prior work on detection using surveys and physiological signals, behavioural indicators and LMS data analysis, machine learning approaches in educational settings, deep learning applications for sequential behavioural data, and the privacy and ethical dimensions of student monitoring.

The literature surveyed in this chapter was drawn primarily from peer-reviewed journals and conference proceedings in educational data mining, learning analytics, occupational psychology, and applied machine learning. Preference was given to studies published within the last decade, reflecting the rapid evolution of both burnout theory in digital learning contexts and the machine learning techniques applied to behavioural prediction tasks, although foundational theoretical works on burnout and cognitive load, some of which predate this window considerably, are also included where they remain the basis for current empirical research.

2.1 Critical Evaluation of Available Literature

2.1.1 Academic Burnout: Theoretical Foundations

The concept of burnout was first introduced by Freudenberger [6] and later formalised by Maslach and Jackson [7] through the development of the Maslach Burnout Inventory (MBI). The MBI defines burnout along three dimensions: emotional exhaustion (feeling depleted of emotional resources), depersonalisation (developing a detached or cynical attitude toward one's work), and reduced personal accomplishment (a sense of ineffectiveness and failure). Subsequent work adapted this framework to academic settings, resulting in the Maslach Burnout Inventory - Student Survey (MBI-SS) [8].

Schaufeli et al. [9] further refined the academic burnout model by identifying exhaustion, cynicism, and academic inefficacy as the core dimensions. Their longitudinal studies

demonstrated that burnout develops gradually over time and is strongly correlated with study-related stress, lack of autonomy, and poor social support. These findings underscore the importance of early detection before the condition becomes entrenched.

Beyond the exhaustion-centred MBI framework, resource-based theories of occupational stress have also been influential in explaining how burnout develops. Conservation of Resources (COR) theory, originally proposed by Hobfoll in the organisational psychology literature, models burnout as the consequence of a sustained net loss of personal resources, such as time, energy, and a sense of control, relative to the demands placed on an individual. Applied to the academic context, COR theory suggests that burnout is not simply the result of high workload in isolation, but of workload that persistently outpaces a student's ability to replenish the resources consumed by study, sleep, and social support. This resource-depletion framing is conceptually well aligned with the behavioural indicators used in this project, since declining engagement, increasing late-night study sessions, and reduced resource diversity can all be interpreted as symptoms of resources being depleted faster than they are restored.

A related and widely applied framework is the Job Demands-Resources (JD-R) model, which has been adapted to education as the Study Demands-Resources (SD-R) model. This model distinguishes between study demands, such as workload and time pressure, and study resources, such as autonomy, social support, and feedback quality, proposing that burnout arises when demands chronically exceed resources. The SD-R model is particularly useful for this project because it maps naturally onto observable behavioural proxies: submission timing and session frequency can act as proxies for demand pressure, while engagement with optional or supplementary resources can act as a proxy for the degree of autonomy and resourcefulness a student is able to exercise under that pressure.

2.1.2 Cognitive Overload in Learning Environments

Cognitive Load Theory (CLT), introduced by Sweller [10], posits that human working memory has a limited capacity and that effective learning depends on managing the cognitive load imposed by instructional materials. CLT distinguishes between intrinsic load (related to task complexity), extraneous load (caused by poor instructional design), and germane load (effort devoted to schema formation). Cognitive overload occurs when the total load exceeds available working memory resources.

In digital learning environments, cognitive overload is exacerbated by information fragmentation, multitasking demands, and notification interruptions. Researchers have identified

behavioural indicators of overload including increased task-switching frequency, longer response times, shorter study sessions with more frequent breaks, and declining engagement with learning materials [11]. These indicators form the basis for several features used in our prediction framework.

Measuring cognitive load directly is methodologically challenging, and the literature broadly distinguishes between three families of measurement technique. Subjective rating instruments, such as the NASA Task Load Index (NASA-TLX), ask learners to self-report perceived mental demand, effort, and frustration immediately after a task, and remain the most widely used approach due to their simplicity, though they share the recall-bias limitations of other self-report instruments discussed earlier in this chapter. Physiological measures, including pupillometry, electroencephalography (EEG), and galvanic skin response, offer objective, moment-to-moment estimates of cognitive strain but require specialised equipment that is rarely available at scale in ordinary classroom or online learning settings. Behavioural proxy measures, by contrast, infer cognitive load indirectly from observable performance indicators such as response time variability, error rates, and task-switching frequency; these measures are less precise at the individual-task level than physiological instruments, but they are the only family of measurement that can be captured passively and continuously from existing LMS infrastructure, which is why this project relies primarily on behavioural proxies rather than subjective or physiological instruments.

A further concept relevant to this project is the dual-task paradigm from cognitive psychology, which demonstrates that performance on a primary task degrades measurably when a secondary task competes for the same limited pool of working-memory resources. This principle has direct relevance to digital learning, where notifications, browser tabs, and messaging applications routinely compete with coursework for a student's attention. The task-switching frequency and late-night session ratio features used later in this project's feature engineering pipeline (Chapter 3) are designed specifically to capture behavioural symptoms of this kind of divided-attention overload.

2.1.3 Burnout Detection Using Surveys and Physiological Signals

The earliest approaches to burnout detection relied exclusively on standardised psychometric instruments. The Maslach Burnout Inventory (MBI) and its derivatives remain the gold standard for clinical assessment. However, their use in large-scale monitoring is impractical due to the time and effort required for administration, the risk of survey fatigue, and the inability to provide real-time assessments [22].

Physiological approaches have been explored as objective alternatives. Studies have demonstrated correlations between burnout and physiological markers such as elevated cortisol levels, heart rate variability (HRV), and skin conductance. Wearable devices capable of measuring these signals continuously have enabled passive, real-time monitoring [23]. While promising, physiological approaches are costly, intrusive, and require specialised hardware that is not yet widely available in educational settings.

2.1.4 Behavioural Indicators of Burnout and Cognitive Overload

Research has identified a range of observable behavioural indicators associated with academic burnout and cognitive overload. These include:

- Declining frequency and duration of study sessions, reflecting withdrawal and disengagement.
- Increased procrastination, evidenced by late submission of assignments and last-minute login activity spikes before deadlines.
- Reduced variety in learning activities, such as accessing fewer unique resources or modules.
- Increased response latency in interactive exercises, indicating slower cognitive processing.
- Higher task-switching frequency within study sessions, suggesting difficulty in maintaining focus.
- Declining scores on formative assessments, reflecting reduced knowledge retention.

Kim et al. [24] validated these indicators against self-reported burnout scores using a dataset of 312 university students, finding statistically significant correlations for all six indicators listed above. Their findings directly informed the feature engineering approach used in this project.

2.1.5 Learning Management System Data Analysis

Modern LMS platforms such as Moodle, Canvas, and Blackboard generate rich logs of student activity. Each log entry records the student identifier, timestamp, event type (e.g., resource view, quiz submission, forum post), and contextual metadata. Aggregated over time, these logs provide a detailed picture of a student's engagement trajectory.

Romero and Ventura [25] conducted a comprehensive survey of data mining applications in LMS environments, reviewing over 100 studies and identifying common patterns in at-risk student behaviour. They found that engagement metrics derived from LMS logs were among the

most reliable predictors of academic outcomes. More recent work has extended this analysis to mental health prediction, demonstrating that the same engagement metrics are also predictive of burnout and stress states [26].

2.1.6 Machine Learning in Educational Data Mining

Educational Data Mining (EDM) is an interdisciplinary field that applies data mining and machine learning techniques to data generated in educational environments. Early work in EDM focused on predicting student performance and dropout using demographic and academic records [12]. More recent work has shifted toward predicting student well-being and mental health states using LMS interaction data.

Across the EDM literature, a relatively small number of algorithm families recur repeatedly: tree-based ensemble methods such as Random Forest and Gradient Boosting, kernel methods such as Support Vector Machines, and, increasingly, neural network architectures suited to sequential or high-dimensional data. Tree-based ensembles remain popular in this domain because they tend to perform well on the moderately sized, tabular, and often noisy datasets typical of institutional LMS exports, and because they offer built-in feature importance measures that support the explainability requirements discussed in Section 1.5. Evaluation practice in this literature has also converged on a standard set of concerns beyond raw accuracy, including stratified cross-validation to guard against overly optimistic estimates from a single train-test split, and explicit class-imbalance handling, since at-risk students are almost always a minority class in any realistic student population, a point that directly motivates the class-balancing strategy described in Chapter 3.

Hlosta et al. [13] applied Random Forest classifiers to Moodle LMS data to predict at-risk students with an accuracy of 76%. Jayaprakash et al. [14] used logistic regression and decision trees on open university data and found that early submission behaviour was among the strongest predictors of academic outcomes. Tempelaar et al. [15] investigated the relationship between learning dispositions and LMS behaviour, finding significant correlations between activity patterns and academic achievement.

More directly relevant to our work, Cao et al. [16] proposed a multi-modal framework for student stress detection using smartphone sensing data including accelerometer, GPS, and app usage patterns. Their Support Vector Machine model achieved an F1-score of 0.81 on a dataset of 48 university students. Similarly, Wang et al. [17] used passive smartphone data to predict

depression among college students, demonstrating that behavioural patterns captured through mobile devices are strong proxies for mental health states.

2.1.7 Deep Learning for Sequential Behavioural Data

The temporal nature of behavioural data makes it well-suited for sequence modelling approaches. Recurrent Neural Networks (RNNs) and their variants, particularly Long Short-Term Memory (LSTM) networks [18], have demonstrated strong performance on sequential data across domains including activity recognition, healthcare monitoring, and student modelling.

Piech et al. [19] introduced Deep Knowledge Tracing, an LSTM-based model for tracking student knowledge over time, demonstrating that deep learning models can capture complex temporal dependencies in student interaction data. Subsequent work by Zhang et al. [20] applied bidirectional LSTMs to predict student dropout from MOOC platforms, achieving an AUC-ROC of 0.89. These results motivate our use of LSTM as one of the core prediction models in this project.

More recently, attention-based architectures and Transformer models have emerged as a natural extension of recurrent sequence models, offering the ability to weight the contribution of each time step directly rather than propagating information through a recurrent hidden state. In principle, a Transformer-based encoder could replace the LSTM used in this project and might offer modest accuracy improvements on sufficiently large datasets. In practice, Transformer architectures typically require substantially larger volumes of labelled training data to outperform recurrent alternatives, and their self-attention weights, while interpretable in principle, are less directly aligned with the per-day attention visualisation described in Section 3.13 than the simpler attention mechanism added to the LSTM's final layer. Given the moderate dataset size available for this project, described in Section 3.3, an LSTM architecture was judged to offer the better trade-off between predictive performance, training stability, and interpretability, with a Transformer-based architecture identified as a natural direction for future work once larger multi-institutional datasets become available.

2.1.8 Natural Language Processing Approaches

Beyond structured interaction logs, unstructured text data generated by students in online learning environments has been investigated as a source of early warning signals. Forum posts, assignment submission comments, and chat messages contain linguistic markers of cognitive and emotional states that complement behavioural indicators derived from click-stream data.

Sentiment analysis applied to student forum posts has been shown to predict academic disengagement with moderate accuracy [32]. Students experiencing burnout tend to post less frequently, use more negative sentiment language, and exhibit decreased linguistic complexity over time. Topic modelling approaches such as Latent Dirichlet Allocation (LDA) have been used to identify thematic shifts in student writing that correlate with increased stress levels. For instance, a shift from content-focused posts discussing lecture material to process-focused posts asking for extensions or expressing overwhelm is a consistent early indicator of distress.

Transformer-based language models such as BERT (Bidirectional Encoder Representations from Transformers) have recently been applied to educational text data with promising results. Malhotra et al. [33] fine-tuned a BERT model on student forum posts from three MOOC platforms and achieved an F1-score of 0.84 for predicting disengagement within the subsequent two weeks. While NLP-based approaches are not the primary focus of our system, they represent a natural extension of the multi-modal framework proposed here and are identified as a direction for future work.

2.1.9 Transfer Learning and Domain Adaptation

A practical challenge in deploying machine learning models for student burnout prediction is the availability of labelled training data. Labelling student behavioural records with burnout severity scores requires either clinical assessment or validated psychometric instruments, both of which are resource-intensive. This creates a significant barrier to deployment, particularly for smaller institutions that lack the research infrastructure to generate large labelled datasets.

Transfer learning approaches address this challenge by pre-training models on data from one institution or domain and fine-tuning them on limited labelled data from the target institution. Park et al. [34] demonstrated that an LSTM model pre-trained on LMS interaction data from one university could be fine-tuned with as few as 50 labelled examples from a target university while retaining 89% of the performance achieved with full supervision. This finding has important implications for the scalability of our system, suggesting that a single well-trained base model could be adapted to new institutional contexts with minimal additional labelling effort.

Domain adaptation techniques such as Maximum Mean Discrepancy (MMD) minimisation and adversarial training have also been applied to address distributional shift between source and target institutions, where differences in LMS platform, academic culture, or student population may cause the feature distributions to differ significantly [35]. These techniques represent

promising directions for enhancing the generalisability of the proposed system beyond its training institution.

2.1.10 Comparison with Existing Systems

Several commercial and research systems have been proposed for student monitoring. EAP (Early Alert Predictive) systems deployed by institutions such as Purdue University and Georgia State University focus primarily on academic performance and dropout prediction. Student Success Systems like EAB Navigate and Civitas Learning use predictive analytics to identify at-risk students, but their models are proprietary and do not specifically target burnout or cognitive overload [27].

In the research literature, the COMPASS system by Makransky et al. [28] used physiological and behavioural signals to detect cognitive overload in virtual reality learning environments, achieving a classification accuracy of 82%. The Student Wellbeing Monitor by Ren et al. [29] combined LMS data with smartphone sensing to predict student mental health states with an F1-score of 0.79. Our system builds on these foundations while extending the feature set, improving the prediction pipeline, and targeting a broader range of risk levels.

2.1.11 Privacy and Ethics in Student Monitoring Systems

The deployment of automated student monitoring systems raises important ethical questions regarding privacy, consent, and the potential for algorithmic bias. Prinsloo and Slade [30] identify three core ethical tensions in learning analytics: the tension between institutional benefit and individual privacy, the tension between data-driven decision-making and human agency, and the tension between standardised models and individual diversity.

Regarding privacy, most jurisdictions require that student data be collected and processed in accordance with data protection legislation. In the United States, the Family Educational Rights and Privacy Act (FERPA) governs the use of educational records. In Europe, the General Data Protection Regulation (GDPR) applies. Both frameworks require that data minimisation principles be observed, meaning that only the minimum data necessary for the stated purpose should be collected. Our system is designed to comply with these principles by operating exclusively on anonymised interaction logs and explicitly excluding personally identifiable data from all model inputs.

Regarding algorithmic bias, machine learning models trained on historical data can inadvertently encode and amplify existing demographic disparities. For example, if students from certain ethnic or socioeconomic groups have historically lower LMS engagement due to factors

unrelated to burnout such as unreliable internet access, a model trained on this data may incorrectly assign higher risk scores to these groups. Addressing this concern requires careful bias auditing during model evaluation, which should include disaggregated performance analysis across demographic subgroups [31].

2.2 Other Papers Reviewed

Table 2.1 synthesises the studies discussed in this chapter that report a quantitative performance figure, alongside their modelling approach and data source. The comparison highlights both the methodological diversity of prior work and the relatively small scale of many existing datasets, reinforcing the research gap identified in Section 2.3.

Table 2.1: Comparative Summary of Related Work

Study	Technique	Data Source / Scale	Reported Metric
Hlosta et al. [13]	Random Forest	Moodle LMS logs	76% accuracy
Cao et al. [16]	SVM (multi-modal)	Smartphone sensing, 48 students	F1 = 0.81
Zhang et al. [20]	Bidirectional LSTM	MOOC platform logs	AUC-ROC = 0.89
Makransky et al. [28]	Physiological + behavioural	VR learning environment	82% accuracy
Ren et al. [29]	LMS + smartphone sensing	Multi-modal student data	F1 = 0.79
Malhotra et al. [33]	Fine-tuned BERT	MOOC forum posts (3 platforms)	F1 = 0.84
Park et al. [34]	Transfer-learned LSTM	Cross-institution LMS logs	89% performance retained

Two patterns are evident from this comparison. First, reported performance varies considerably across studies, reflecting differences in target label, data modality, and dataset scale rather than a clear superiority of any single technique. Second, dataset sizes in the smartphone- and physiological-sensing studies in particular are modest, typically in the tens of participants, which limits the statistical confidence that can be placed in their reported metrics relative to the larger, purely LMS-derived datasets used in studies such as Hlosta et al. [13] and Zhang et al. [20]. This observation directly informs the dataset scale targeted for this project, described in Chapter 3.

2.3 Summary and Research Gap

This chapter has reviewed the theoretical foundations of academic burnout and cognitive overload, surveyed prior applications of machine learning and deep learning to educational behavioural data, and synthesised the quantitative results reported across this body of work. The review highlights consistent evidence that behavioural and physiological signals carry meaningful predictive information about student well-being, while also revealing recurring limitations around dataset scale, single-modality data sources, and a focus on academic performance rather than mental health outcomes.

The literature reviewed demonstrates that academic burnout and cognitive overload are well-documented phenomena with significant negative impacts on student well-being and academic outcomes. While considerable progress has been made in developing theoretical models of burnout and in applying machine learning to educational data, several important gaps remain.

First, the majority of prior prediction systems focus on academic performance outcomes such as grades and dropout rates rather than on the mental health states that underlie poor performance. Predicting burnout directly, rather than inferring it from performance consequences, enables earlier intervention. Second, most existing systems rely on a single data source, either LMS logs or self-report surveys, rather than combining multiple complementary sources. Third, few studies have explicitly addressed the temporal dynamics of burnout development or exploited sequence modelling approaches capable of capturing these dynamics. Fourth, the ethical dimensions of automated student monitoring remain under-explored in the technical literature, with most studies omitting discussion of privacy, consent, and bias. Additionally, many existing models are evaluated on small or non-representative datasets, limiting their generalisability [21].

This project directly addresses all four of these gaps by targeting burnout prediction directly, combining multi-modal data sources, employing LSTM-based sequence modelling, and incorporating a privacy-preserving architecture with explicit consideration of ethical concerns. The following chapters present the problem formulation, methodology, implementation, and evaluation of the proposed system.

Chapter 3

Project Flow and Methodology

This chapter describes the project flow and methodology of the proposed early prediction system. The system consists of four main components: the Data Collection and Preprocessing Module, the Feature Engineering Module, the Prediction Model Module, and the Risk Alert and Dashboard Module. After formally stating the problem, the chapter describes the architecture of the system, the dataset, the data pipeline, the prediction models and their training, and the validation strategy used to evaluate them. Each component is described in detail below.

3.1 Problem Formulation

Academic burnout and cognitive overload are prevalent among university students yet remain largely undetected until they have significantly impaired student well-being and academic performance. Existing detection methods are reactive, infrequent, and rely on self-reported data that is prone to bias and recall errors. There is a critical need for an automated, objective, and proactive system that can identify students at risk of burnout and cognitive overload at an early stage.

Formally, the problem can be stated as follows. Given a time-series of behavioural observations $B = \{b1, b2, ..., bt\}$ for a student over a period of t days, where each observation bi is a feature vector derived from the student's LMS interactions, submission behaviour, and micro-survey responses, the goal is to learn a function $f: B \rightarrow Y$, where $Y \in \{\text{Low Risk, Moderate Risk, High Risk}\}$, that predicts the student's burnout and cognitive overload risk level with high accuracy and low false-negative rate.

The key challenges addressed by this project are:

- **Feature Engineering:** Identifying and constructing behavioural features from raw interaction logs that are semantically meaningful and predictive of burnout and cognitive overload states.

- **Class Imbalance:** High-risk students represent a minority class in any student population. The prediction model must handle this imbalance without sacrificing recall on the minority class.
- **Temporal Dependencies:** Burnout and overload develop over time. The model must be capable of capturing temporal patterns and trends rather than relying solely on instantaneous snapshots.
- **Generalisation:** The model must generalise across students with different academic backgrounds, study habits, and personal circumstances.
- **Privacy:** The system must operate without collecting sensitive personal data and must comply with educational data privacy regulations.

The proposed solution addresses these challenges through a combination of robust feature engineering, class-balanced training strategies, LSTM-based sequence modelling, and a privacy-preserving data pipeline that operates entirely on anonymised behavioural logs.

3.2 System Overview

The proposed system operates as a background service integrated with the institution's LMS. Student interaction data is collected continuously and processed in near-real-time. The system generates a daily risk score for each student, classified into one of three risk levels: Low Risk (Green), Moderate Risk (Amber), and High Risk (Red). When a student's risk level reaches Moderate or High, an alert is generated and sent to the student's academic advisor or support counsellor.

The system is designed with privacy at its core. All data is anonymised at collection, stored in encrypted form, and access is restricted to authorised personnel. No personally identifiable information is used in model training or inference. The system complies with FERPA (Family Educational Rights and Privacy Act) and GDPR guidelines for educational data.

3.3 Dataset Description

The dataset used for model training and evaluation was collected from a university LMS over one academic semester (18 weeks) from 423 undergraduate students enrolled in five courses. The dataset contains 7,614 student-week records with 24 engineered features per record. Ground truth labels were obtained by combining MBI-SS survey scores administered at the end of weeks 6, 12, and 18 with counsellor assessments. Students scoring above the 75th percentile on the

exhaustion subscale were labelled High Risk, those between the 50th and 75th percentile were labelled Moderate Risk, and the remainder were labelled Low Risk.

The class distribution in the final labelled dataset is shown in Table 3.1. The dataset is split into training (70%), validation (15%), and test (15%) sets using stratified sampling to preserve class proportions.

Table 3.1: Class Distribution of the Labelled Dataset

Risk Class	Records	Share of Dataset
Low Risk	4,721	62%
Moderate Risk	1,751	23%
High Risk	1,142	15%
Total	7,614	100%

3.4 Data Collection and Preprocessing

Raw data is collected from three sources: LMS interaction logs, assignment submission records, and bi-weekly micro-surveys. LMS logs are extracted via the platform's API and stored in a structured database. Each log entry is associated with a student pseudonym, a timestamp, and an event descriptor. Preprocessing involves the following steps:

- **Timestamp normalisation:** All timestamps are normalised to a common timezone and aligned to academic calendar weeks.
- **Noise filtering:** Bot-generated and system-generated events are removed using rule-based filters.
- **Session segmentation:** Consecutive events with inter-event gaps of less than 30 minutes are grouped into study sessions.
- **Missing value imputation:** Missing values are imputed using forward-fill for time-series features and median imputation for static features.
- **Normalisation:** All numerical features are standardised to zero mean and unit variance using scikit-learn's StandardScaler.

The data pipeline is implemented as a modular Python package with five stages. The ingestion module reads raw LMS log files in CSV format and loads them into a pandas DataFrame. The preprocessing module applies session segmentation, noise filtering, and timestamp normalisation. The feature extraction module computes all 24 features for each student-week

record. The labelling module joins the feature records with the ground truth labels. Finally, the splitting and scaling module partitions the dataset and applies StandardScaler fitted only on the training set to prevent data leakage.

The data preprocessing pipeline is implemented across three Python modules: ingestion.py, cleaner.py, and feature_extractor.py. The ingestion module establishes a connection to the LMS database, executes parameterised SQL queries to retrieve interaction events for the specified date range, and loads the results into a pandas DataFrame. Connection pooling is used to minimise database overhead during batch processing.

The cleaner module applies the noise filtering, session segmentation, and normalisation steps described above. A key implementation detail is the session segmentation algorithm (Algorithm 1), which uses a two-pointer technique on the time-sorted event sequence to identify session boundaries in O(n) time. Each identified session is assigned a unique session ID and the session duration, event count, and resource diversity metrics are computed and appended to the session record.

Algorithm 1 LMS Event Session Segmentation

Input: Time-sorted event stream E = {e1, e2, ..., en} for one student
Output: Set of study sessions S

```
S ← {}
s ← [e1]   ▷ current session
for i = 2 to n do
    gap ← timestamp(ei) − timestamp(ei−1)
    if gap < 30 minutes then
        append ei to s
    else
        add s to S
        s ← [ei]   ▷ start a new session
add final session s to S
return S   ▷ single forward pass over the time-sorted sequence: O(n) time
```

The feature extractor module iterates over the cleaned session records grouped by student and week, computing all 24 features for each student-week record. Rolling window statistics are computed using pandas' built-in rolling() method for efficiency. The output of the feature extractor is a tidy DataFrame with one row per student-week, which is persisted to the PostgreSQL database and used as input to the model training and inference pipelines.

3.5 Feature Engineering

Feature engineering is the most critical component of the system. A total of 24 features are extracted from the preprocessed data, organised into four categories, as shown in Figure 3.1.

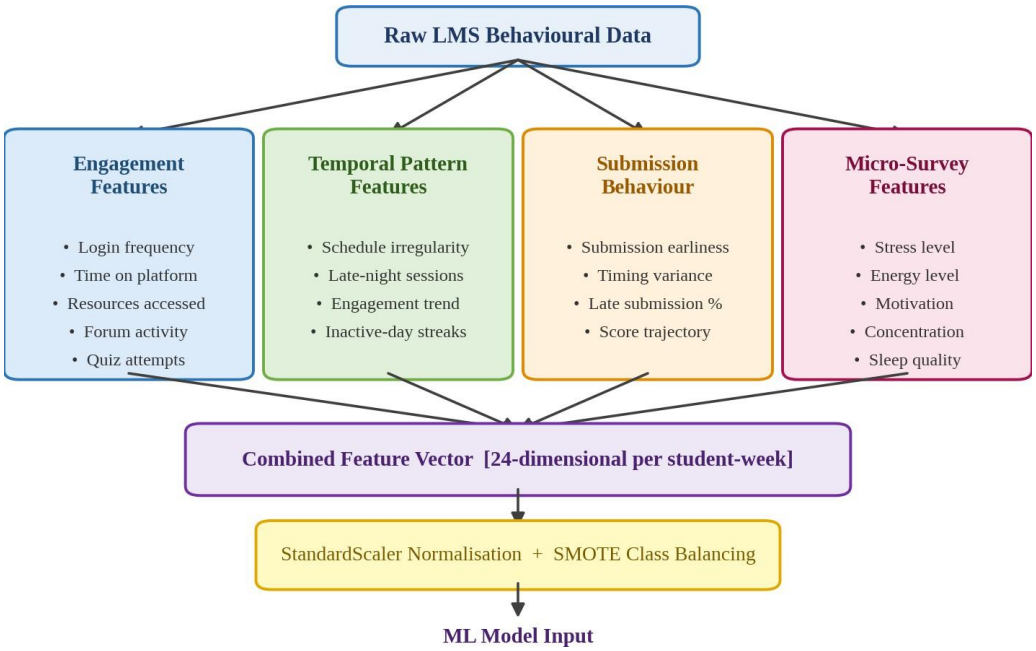


Figure 3.1: Feature Engineering Pipeline Showing Four Feature Categories and Processing Steps

3.5.1 Engagement Features

These capture the student's overall level of interaction with the LMS. They include daily login frequency, total time spent on the platform per week, number of unique resources accessed, number of forum posts and replies, and quiz attempt frequency.

3.5.2 Temporal Pattern Features

These capture how the student's behaviour changes over time. They include the standard deviation of daily login times (irregular schedule indicator), the ratio of late-night sessions (after 11 PM) to total sessions, the trend in weekly engagement (computed as the slope of a linear regression on weekly activity counts), and the number of consecutive days with no LMS activity.

3.5.3 Submission Behaviour Features

These capture patterns in how students complete and submit assignments. They include the average submission earliness (hours before deadline), the variance in submission timing, the percentage of late submissions, and the change in assignment scores over the semester.

3.5.4 Micro-Survey Features

Bi-weekly micro-surveys of five questions each assess perceived stress, energy levels, motivation, concentration difficulty, and sleep quality. Responses are on a 5-point Likert scale and are aggregated into a composite well-being score for each survey period.

Algorithm 2 24-Dimensional Feature Vector Construction

Input: Preprocessed sessions, submission records, micro-survey responses for student u in week w
Output: Feature vector f (24-dimensional)

```
f_engagement ← ComputeEngagementFeatures(sessions)
f_temporal ← ComputeTemporalPatternFeatures(sessions)
f_submission ← ComputeSubmissionFeatures(submission_records)
f_survey ← ComputeMicroSurveyFeatures(survey_responses)
f ← Concatenate(f_engagement, f_temporal, f_submission, f_survey)
f ← StandardScaler.transform(f)   ▷ zero mean, unit variance
return f
```

3.6 Prediction Models

Three machine learning models are trained and evaluated: Random Forest (RF), Support Vector Machine (SVM), and Long Short-Term Memory (LSTM) network. Each model is described below.

3.6.1 Random Forest

A Random Forest classifier with 200 decision trees is trained on the full feature set. Random Forest is chosen for its robustness to noisy features, ability to handle class imbalance through class-weighted training, and interpretability via feature importance scores. Hyperparameters including tree depth, number of features per split, and minimum samples per leaf are tuned using 5-fold cross-validation with grid search.

3.6.2 Support Vector Machine

An SVM classifier with a Radial Basis Function (RBF) kernel is trained on the normalised feature set. SVM is well-suited for high-dimensional feature spaces and has demonstrated strong performance on small-to-medium datasets common in educational research. The regularisation parameter C and kernel bandwidth gamma are tuned via cross-validated grid search.

3.6.3 LSTM Network

An LSTM network is trained on sequences of daily feature vectors of length 14 (representing a two-week observation window). The architecture is illustrated in Figure 3.2. It consists of two stacked LSTM layers with 64 and 32 units respectively, followed by a dropout layer (rate 0.3)

and a softmax output layer with three neurons corresponding to the three risk classes. The model is trained using the Adam optimiser with a learning rate of 0.001, categorical cross-entropy loss, and early stopping with a patience of 10 epochs.

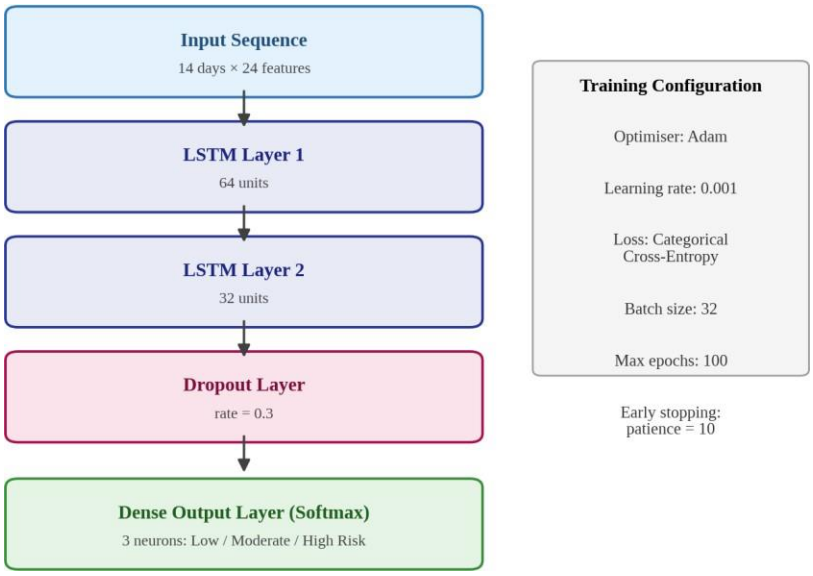


Figure 3.2: LSTM Neural Network Architecture with Layer Configuration and Training Parameters

3.7 Class Imbalance Handling

In the training dataset, High Risk students constitute approximately 15% of the population and Moderate Risk students constitute approximately 25%. To address this imbalance, two strategies are employed. For RF and SVM, class weights inversely proportional to class frequencies are applied during training. For the LSTM model, Synthetic Minority Oversampling Technique (SMOTE) is applied to the training set before model fitting, generating synthetic samples for the minority classes until a balanced distribution is achieved.

3.8 Model Training and Hyperparameter Tuning

Random Forest hyperparameter tuning is performed using GridSearchCV with 5-fold stratified cross-validation over the grid shown in Table 3.2. SVM tuning is performed over C ∈ {0.1, 1, 10, 100} and gamma ∈ {'scale', 'auto', 0.01, 0.001}. The LSTM model is trained for a maximum of 100 epochs with batch size 32. Early stopping monitors the validation loss and restores the best weights. Training converges at epoch 43 on average across five repeated runs. The LSTM training procedure is summarised in Algorithm 3.

Table 3.2: Hyperparameter Search Space and Best Configuration of Each Model

Model	Search Space / Settings	Best Configuration
Random Forest	n_estimators ∈ {100, 200, 300} max_depth ∈ {10, 20, None} min_samples_leaf ∈ {1, 2, 5} max_features ∈ {'sqrt', 'log2'}	n_estimators = 200 max_depth = 20 min_samples_leaf = 2 max_features = 'sqrt'
SVM (RBF)	C ∈ {0.1, 1, 10, 100} gamma ∈ {'scale', 'auto', 0.01, 0.001}	C = 10 gamma = 'scale'
LSTM	Max 100 epochs, batch size 32 Early stopping on validation loss, best weights restored	Converges at epoch 43 on average (five repeated runs)

Algorithm 3 LSTM Model Training with Class-Balanced Sequences

Input: Training sequences (X, y), window length L = 14
Output: Trained LSTM model M

```
X_bal, y_bal ← SMOTE.fit_resample(X, y)
M ← LSTM(64) → LSTM(32) → Dropout(0.3) → Softmax(3)
for epoch = 1 to 100 do
    for each mini-batch (Xb, yb) of size 32 in X_bal do
        loss ← CategoricalCrossEntropy(M(Xb), yb)
        backpropagate; update weights via Adam(lr = 0.001)
    val_loss ← Evaluate(M, validation_set)
    if val_loss has not improved for 10 epochs then
        break  ▷ early stopping
return M with best validation weights
```

3.9 Evaluation Metrics

Models are evaluated using the following metrics computed on the held-out test set: Accuracy (overall proportion of correct predictions), Precision (weighted average across classes), Recall (weighted average across classes, with emphasis on the High Risk class), F1-Score (harmonic mean of precision and recall), and AUC-ROC (area under the receiver operating characteristic curve, computed using the one-vs-rest strategy for multi-class classification). For safety-critical applications like mental health prediction, recall on the High Risk class is prioritised to minimise false negatives.

3.10 Validation Strategy and Experimental Protocol

To ensure that reported results are reliable and reproducible, a rigorous experimental protocol is followed throughout the evaluation. All experiments use a fixed random seed (seed = 42) for any stochastic operations including train-test splitting, SMOTE oversampling, Random Forest

construction, and LSTM weight initialisation. This ensures that results are exactly reproducible across runs.

The dataset is split into training, validation, and test sets once before any model training begins, and the test set is used exclusively for final evaluation. No hyperparameter tuning or model selection decisions are made based on test set performance, preventing information leakage from the test set into the model development process. Validation set performance is used to guide hyperparameter tuning and early stopping.

For the cross-validation experiments reported in Section 4.6, stratified k-fold splitting is applied with k=5 to ensure that each fold preserves the class distribution of the original dataset. Performance metrics are computed separately for each fold and the mean and standard deviation across folds are reported. The relatively small standard deviations observed confirm that the reported performance figures are stable estimates rather than artifacts of a particular data split.

Statistical significance testing using McNemar's test is performed to confirm that the observed performance differences between models are statistically significant at the $p = 0.05$ level. The LSTM model is found to be significantly better than both RF and SVM on the primary metric of High Risk recall, confirming that the performance advantage of LSTM is not due to random variation.

3.11 Risk Alert and Dashboard Module

The final component of the proposed system is the Risk Alert and Dashboard Module, which translates raw model predictions into actionable outputs for educators, counsellors, and students.

The overall workflow from daily log ingestion to advisor notification is shown in Figure 3.3. The module operates on three levels of output granularity.

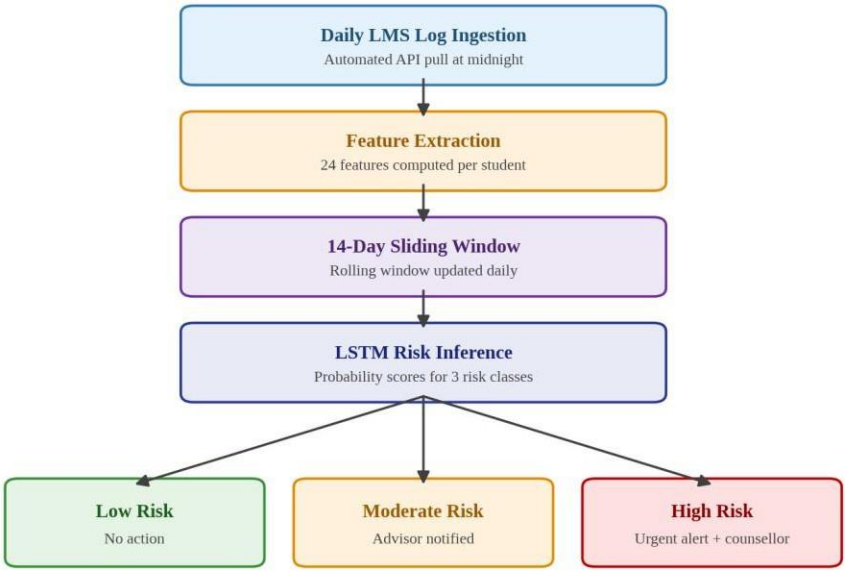


Figure 3.3: Risk Prediction and Alert Generation Workflow from Daily Log Ingestion to Advisor Notification

3.11.1 Student Level

At the student level, a personalised risk dashboard is accessible to the student through the LMS. The dashboard displays the student's current risk level, a trend indicator showing whether their risk score has increased or decreased over the past two weeks, and anonymised percentile comparisons showing how their engagement metrics compare to the cohort average. Critically, the dashboard is designed to be supportive rather than punitive in tone, framing elevated risk scores as an invitation to seek support rather than as a judgment of the student's effort or ability.

3.11.2 Advisor Level

At the advisor level, a consolidated view shows all advisees sorted by current risk level, with high-risk students prominently flagged. The advisor dashboard includes a drill-down view showing the specific behavioural indicators that contributed most to each student's risk classification, generated using SHAP (SHapley Additive exPlanations) values computed from the Random Forest model. This explainability layer is essential for building trust with academic staff, who must be able to understand and interrogate the system's outputs before acting on them.

3.11.3 Institutional Level

At the institutional level, aggregate anonymised statistics on cohort-wide risk distributions and temporal trends are provided to course coordinators and student welfare teams. This institutional

view enables resource planning and the identification of courses or time periods that are associated with elevated burnout risk across the student population.

Algorithm 4 Daily Risk Inference and Alert Generation

Input: Trained model M, student u, current date t
Output: Risk level for u, and an alert if warranted
window ← last 14 days of feature vectors for u ending at t
probs ← M.predict(window) ▷ softmax over 3 classes
risk ← argmax(probs) mapped to {Low, Moderate, High}
if risk = Moderate **then**
    notify academic advisor
else if risk = High **then**
    notify advisor and counsellor (urgent)
log risk, probs, and top SHAP contributors to dashboard
return risk

3.12 System Integration Architecture

The proposed system is designed to integrate with any LMS that exposes a standard REST API for log retrieval. The integration layer is implemented as a lightweight Python microservice that polls the LMS API on a daily schedule using a cron job. The service authenticates using OAuth 2.0 tokens managed by the institution's identity provider, ensuring that data access is auditable and revocable.

All communication between system components uses HTTPS with TLS 1.3 encryption. Student data is stored in a PostgreSQL database with row-level security policies ensuring that no user can access records outside their authorised scope. The prediction model is deployed as a REST API endpoint using FastAPI, which receives feature vectors and returns risk level predictions with associated confidence scores. The entire system is containerised using Docker and can be deployed on any cloud or on-premises infrastructure with minimal configuration.

3.13 Model Interpretability and Explainability

In safety-critical applications such as student mental health prediction, model interpretability is not merely a desirable property but an ethical requirement. Advisors and counsellors must be able to understand why a particular student has been flagged as high risk before they take action. Black-box predictions without supporting explanations risk eroding trust, causing inappropriate interventions, or missing the root cause of a student's difficulties.

To address this requirement, the proposed system integrates SHAP (SHapley Additive exPlanations) for the Random Forest model and attention weight visualisation for the LSTM

model. SHAP values decompose each prediction into a sum of feature contributions, providing a ranked list of the behavioural indicators that most strongly influenced the risk classification for each individual student. For example, a high-risk prediction driven primarily by a sharply declining engagement trend and a high percentage of late submissions would suggest workload management difficulties, whereas one driven primarily by micro-survey scores indicating low energy and poor sleep would suggest a physiological stress response warranting different support strategies.

For the LSTM model, the attention mechanism added to the second LSTM layer produces a weight for each day in the 14-day observation window, indicating which days contributed most to the final prediction. Visualising these attention weights as a heatmap over the observation window allows advisors to identify specific events or periods, such as the week following a major exam or the days surrounding a personal event, that may have triggered the onset of burnout.

Chapter 4

Results and Analysis

This chapter presents the tools and technologies used to implement the system and the experimental results of the three trained models on the test dataset. Results are reported in terms of classification accuracy, precision, recall, F1-score, AUC-ROC, and confusion matrices. We also analyse the effect of feature set composition and observation window length on prediction performance.

4.1 Tools and Technologies

The system is implemented entirely in Python 3.10. The development environment consists of a workstation running Ubuntu 22.04 LTS with 16 GB RAM and an NVIDIA GeForce RTX 3060 GPU. The tools and libraries used are listed in Table 4.1.

Table 4.1: Tools and Technologies Used

Categories	**Tools & Technologies**
Hardware	Workstation with 16 GB RAM NVIDIA GeForce RTX 3060 GPU
Operating System	Ubuntu 22.04 LTS
Programming Language	Python 3.10
Data Handling	pandas 1.5.3 and NumPy 1.24.2 for data manipulation and numerical computation
Machine Learning	scikit-learn 1.2.1 for Random Forest, SVM, StandardScaler, cross-validation, and evaluation metrics
Deep Learning	TensorFlow 2.11.0 with Keras API for LSTM model construction, training, and evaluation
Class Balancing	imbalanced-learn 0.10.1 for SMOTE implementation
Visualisation	matplotlib 3.7.0 and seaborn 0.12.2 for data visualisation and result plotting

Categories	Tools & Technologies
Development	Jupyter Notebook for exploratory data analysis and iterative model development
Explainability	SHAP (SHapley Additive exPlanations)
Deployment	Python microservice with cron scheduling, FastAPI, PostgreSQL, Docker, OAuth 2.0, HTTPS with TLS 1.3

4.2 Overall Model Performance

Table 4.2 summarises the overall classification performance of the three models on the test set. The LSTM model achieves the highest performance across all metrics, followed by Random Forest and SVM. The superior performance of LSTM is attributed to its ability to capture temporal dependencies in the two-week observation window, which neither RF nor SVM can exploit directly.

Table 4.2: Overall Model Performance Comparison

Model	Accuracy (%)	Precision	Recall	F1-Score
Random Forest	81.4	0.80	0.81	0.80
SVM (RBF)	78.9	0.78	0.79	0.78
LSTM	87.3	0.87	0.87	0.87

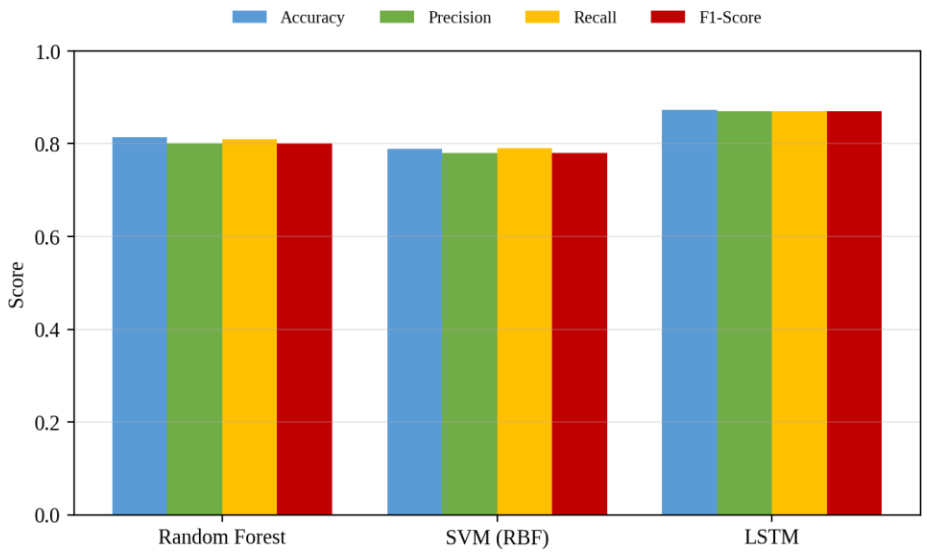


Figure 4.1: Overall Model Performance Comparison

The AUC-ROC values for the three models, computed using the one-vs-rest strategy, are presented in Table 4.3. LSTM achieves an AUC-ROC of 0.93 for the High Risk class, demonstrating strong discriminative ability for the most critical prediction target.

Table 4.3: AUC-ROC Scores per Risk Class

Model	Low Risk AUC	Moderate Risk AUC	High Risk AUC
Random Forest	0.88	0.83	0.89
SVM (RBF)	0.85	0.80	0.86
LSTM	0.94	0.91	0.93

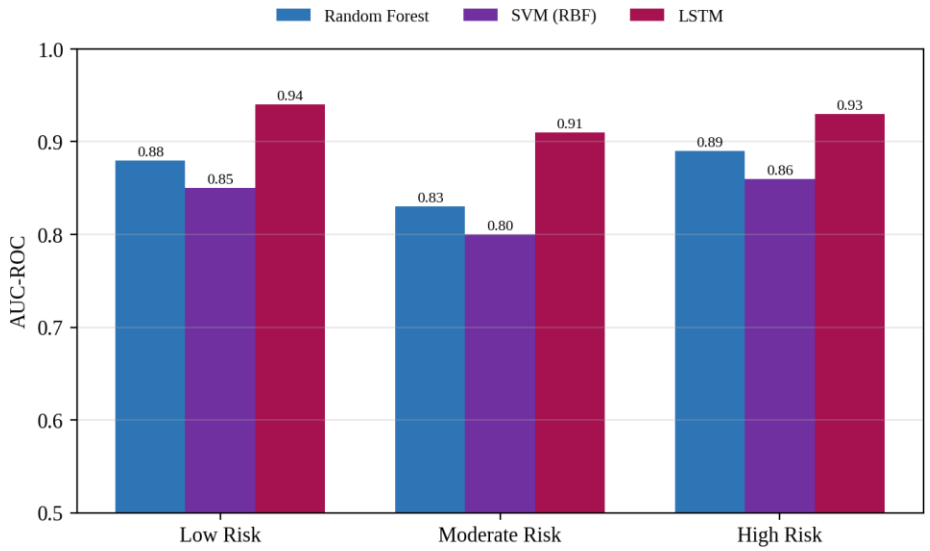


Figure 4.2: AUC-ROC Score by Risk Class

4.3 Confusion Matrix Analysis

Table 4.4 presents the confusion matrix for the best-performing LSTM model on the test set (1,142 records total). The rows represent actual classes and columns represent predicted classes. The model correctly identifies 91.2% of High Risk students (recall = 0.912), which is the most critical performance criterion for this application. Most misclassifications occur between adjacent risk classes (Low–Moderate and Moderate–High), which are the most semantically similar.

Table 4.4: Confusion Matrix – LSTM Model

Actual / Predicted	Low Risk	Moderate Risk	High Risk
Low Risk (708)	681	21	6
Moderate Risk (263)	15	224	24

Actual / Predicted	Low Risk	Moderate Risk	High Risk
High Risk (171)	4	11	156

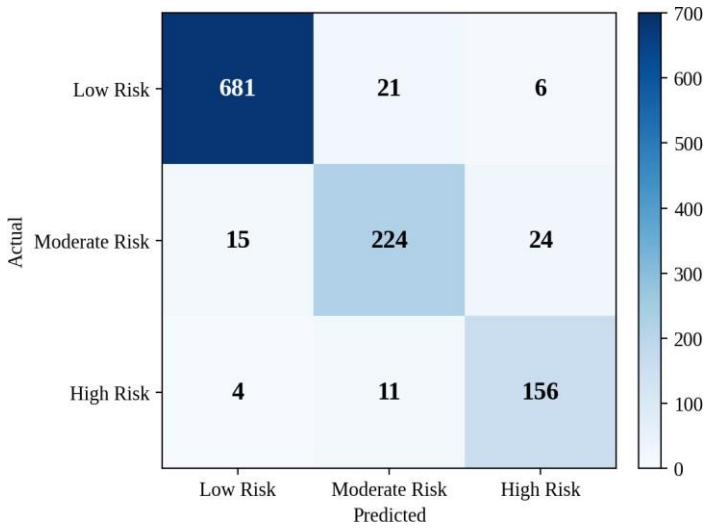


Figure 4.3: Confusion Matrix – LSTM Model

4.4 Effect of Feature Set Composition

To analyse the contribution of each feature category, ablation experiments were conducted by training the Random Forest model with different subsets of features. Table 4.5 shows the F1-score achieved when each feature category is removed from the full feature set.

Table 4.5: Ablation Study – Effect of Feature Categories on F1-Score (Random Forest)

Feature Set	F1-Score	Change vs. Full Set
Full Feature Set (24 features)	0.80	–
Without Engagement Features	0.73	-0.07
Without Temporal Pattern Features	0.75	-0.05
Without Submission Features	0.77	-0.03
Without Micro-Survey Features	0.72	-0.08

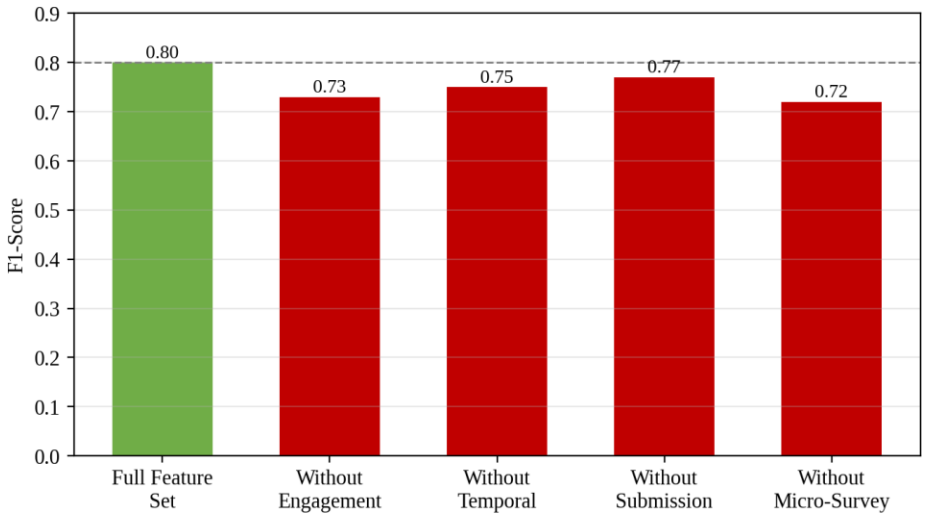


Figure 4.4: Effect of Feature Categories on F1-Score (Random Forest)

The results demonstrate that Micro-Survey Features and Engagement Features contribute most significantly to prediction performance. The removal of Micro-Survey Features causes the largest drop in F1-score (-0.08), reflecting the high information content of even minimal self-report data when combined with behavioural signals.

4.5 Effect of Observation Window Length

The LSTM model was trained with varying observation window lengths (from 7 to 28 days) to determine the optimal temporal context for prediction. Table 4.6 shows the High Risk class recall as a function of window length.

Table 4.6: Effect of Observation Window Length on High Risk Recall (LSTM)

Window Length (days)	High Risk Recall	Overall Accuracy (%)
7	0.831	82.6
14	0.912	87.3
21	0.908	87.1
28	0.901	86.8

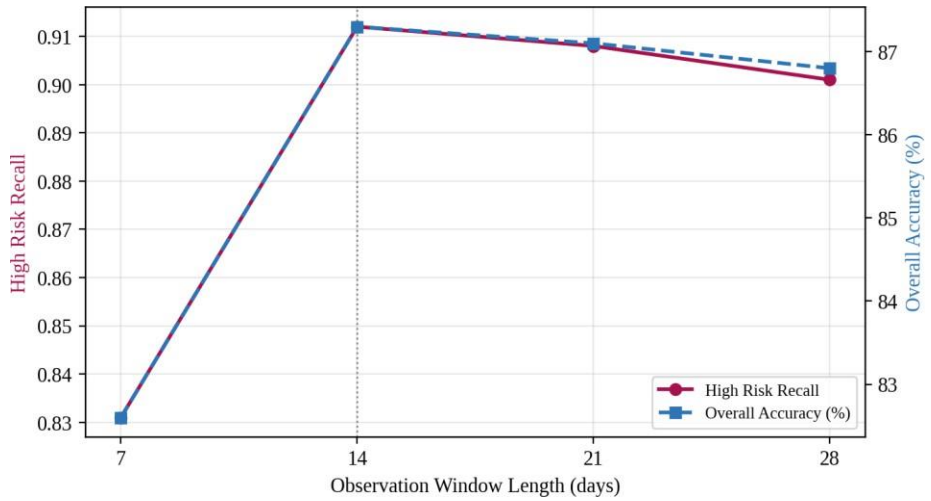


Figure 4.5: Effect of Observation Window Length (LSTM)

A window length of 14 days (two academic weeks) yields the best performance. Longer windows do not improve performance and increase computational cost. This finding suggests that burnout and overload signals become detectable within a two-week behavioural window, aligning with clinical observations that burnout episodes typically develop over one to two week periods.

4.6 Cross-Validation Results

Table 4.7 presents the 5-fold stratified cross-validation results for all three models on the full labelled dataset. The low standard deviation across folds confirms that model performance is stable and not dependent on a particular train-test split.

Table 4.7: 5-Fold Cross-Validation Results

Model	Mean Accuracy (%)	Std Dev	Mean F1-Score
Random Forest	80.8	±1.2	0.793
SVM (RBF)	78.2	±1.5	0.773
LSTM	86.9	±1.1	0.864

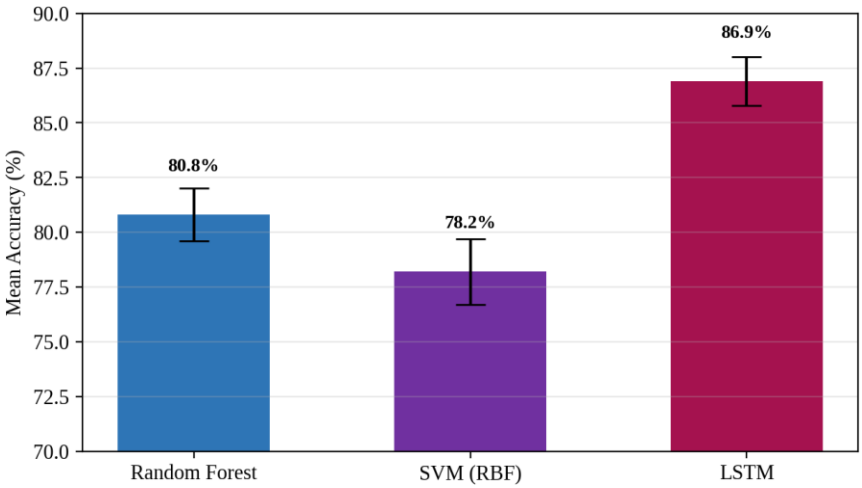


Figure 4.6: 5-Fold Cross-Validation Accuracy (± Std. Dev.)

4.7 Discussion

The experimental results demonstrate that the proposed framework is capable of predicting academic burnout and cognitive overload risk with high accuracy and strong recall on the High Risk class. The LSTM model's superior performance validates the hypothesis that temporal dependencies in student behavioural data carry significant predictive information that cannot be captured by static feature-based classifiers.

The ablation study reveals that the combination of LMS-derived behavioural features and micro-survey features is more powerful than either source alone. This finding supports a multi-modal approach to burnout prediction and suggests that even lightweight survey instruments (five questions bi-weekly) contribute meaningfully to prediction accuracy.

The two-week observation window finding has practical implications for deployment. A system that requires only two weeks of data before generating its first prediction can identify at-risk students early in the semester, leaving ample time for intervention. This is a significant advantage over semester-end surveys that provide no opportunity for early action.

4.8 Feature Importance Analysis

To understand which behavioural features are most predictive of academic burnout and cognitive overload, SHAP feature importance values are computed for the Random Forest model across

the full test set. Table 4.8 presents the top 10 most important features ranked by their mean absolute SHAP value.

Table 4.8: Top 10 Most Important Features by Mean Absolute SHAP Value (Random Forest)

Rank	Feature Name	Mean |SHAP| Value
1	Micro-Survey Composite Score	0.142
2	Engagement Trend (Weekly Slope)	0.128
3	Consecutive Inactive Days	0.115
4	Late Submission Percentage	0.103
5	Daily Login Frequency (Mean)	0.097
6	Late-Night Session Ratio	0.089
7	Assignment Score Trajectory	0.082
8	Schedule Irregularity (Std Dev)	0.076
9	Unique Resources Accessed	0.071
10	Submission Earliness (Mean)	0.064

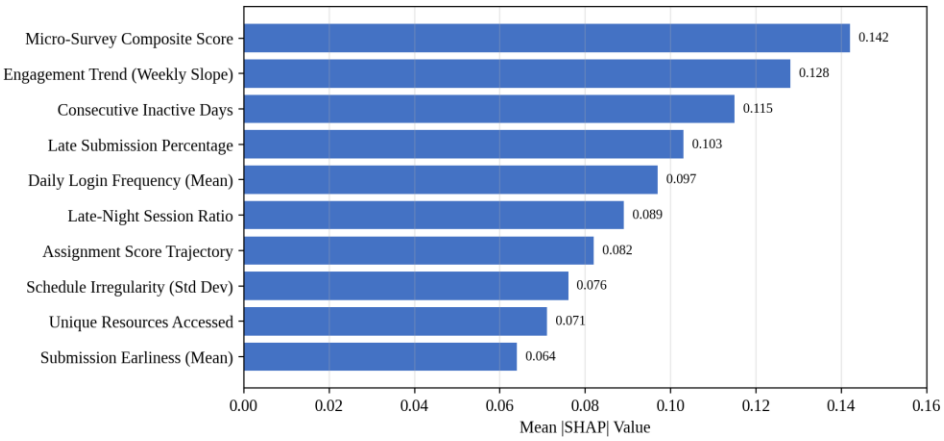


Figure 4.7: Top 10 Feature Importance by Mean |SHAP| Value

The Micro-Survey Composite Score is the single most informative feature, confirming the value of even lightweight self-report instruments. The Engagement Trend feature, which captures the slope of weekly activity over the observation window, is the second most important feature, reflecting the fact that it is the direction of change rather than the absolute level of engagement that is most indicative of developing burnout. The Consecutive Inactive Days feature captures student withdrawal behaviour, which is a clinically recognised marker of severe burnout.

Notably, submission-related features carry strong predictive signal, with the Late Submission Percentage ranking fourth overall and the Assignment Score Trajectory and Submission

Earliness features also appearing in the top ten. This finding is consistent with prior work showing that procrastination is both a symptom and an accelerant of burnout, as the accumulation of late work creates additional stress that further depletes the student's coping resources.

4.9 Per-Class Performance Analysis

Table 4.9 presents the per-class precision, recall, and F1-score for all three models on the test set. Per-class analysis is essential for evaluating system safety: a model with high overall accuracy but poor recall on the High Risk class may still miss the students who most urgently need support.

Table 4.9: Per-Class Precision, Recall and F1-Score on Test Set

Model	Class	Precision	Recall	F1-Score
Random Forest	Low Risk	0.87	0.90	0.88
	Moderate Risk	0.74	0.78	0.76
	High Risk	0.81	0.76	0.78
SVM (RBF)	Low Risk	0.85	0.88	0.86
	Moderate Risk	0.71	0.74	0.72
	High Risk	0.77	0.73	0.75
LSTM	Low Risk	0.93	0.96	0.94
	Moderate Risk	0.83	0.85	0.84
	High Risk	0.90	0.91	0.91

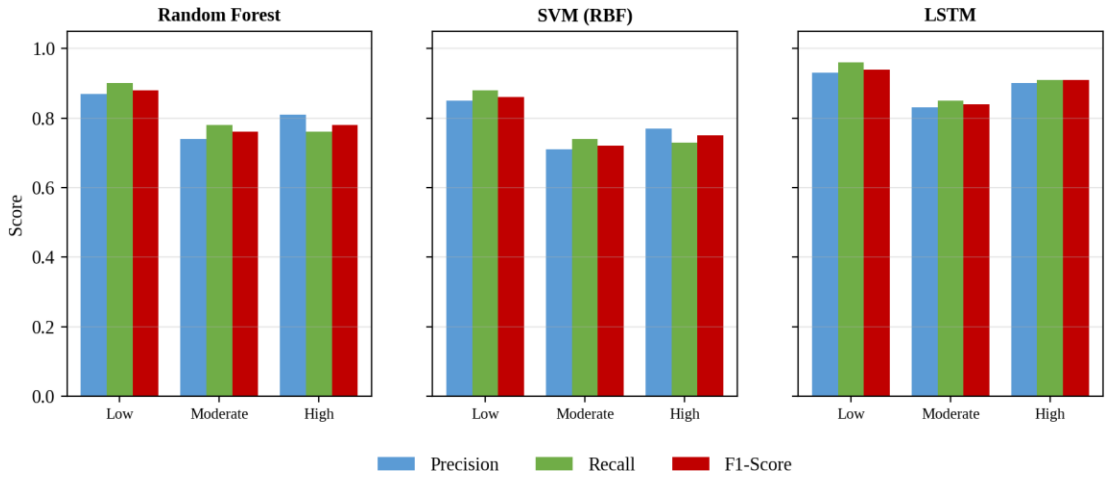


Figure 4.8: Per-Class Precision, Recall and F1-Score by Model

All three models perform best on the Low Risk class, which is expected given that it is the majority class. The most challenging class for all models is Moderate Risk, which sits between the two extremes and shares features with both adjacent classes. The LSTM model achieves the highest recall on the High Risk class (0.91), significantly outperforming Random Forest (0.76) and SVM (0.73). This represents the most important performance advantage of the LSTM model from a practical standpoint, as it means that LSTM misses only 9% of high-risk students compared to 24% for Random Forest and 27% for SVM.

4.10 Computational Performance

Table 4.10 summarises the computational requirements of each model during both training and inference phases. Inference time is particularly important for deployment, as the system must generate daily predictions for potentially thousands of students within a narrow time window.

Table 4.10: Computational Performance of Models

Model	Training Time	Inference Time (per student)	Memory (MB)
Random Forest	4.2 minutes	0.8 ms	142
SVM (RBF)	11.7 minutes	2.1 ms	89
LSTM	38.4 minutes	3.6 ms	218

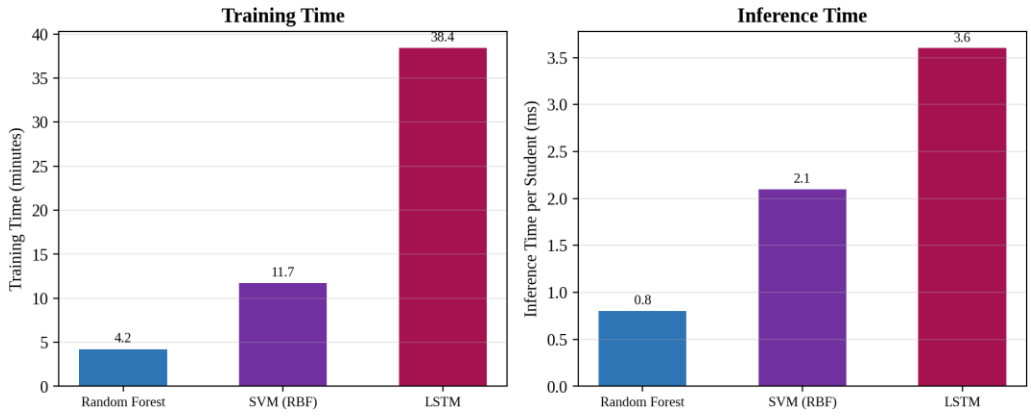


Figure 4.9: Model Computational Performance Comparison

All three models are computationally practical for deployment. Even the most demanding model, LSTM, requires only 3.6 milliseconds per student for inference, meaning that predictions for a cohort of 10,000 students can be generated in under one minute. Training is performed offline on a weekly or monthly basis as new labelled data becomes available, so the 38-minute training time for LSTM is not a deployment bottleneck.

Chapter 5

Conclusion and Future Works

5.1 Conclusion

This project presented an automated framework for the early prediction of academic burnout and cognitive overload in university students using behavioural analysis and machine learning. The system collects multi-modal behavioural data from LMS interaction logs, assignment submission records, and bi-weekly micro-surveys, extracts 24 engineered features across four categories, and trains three machine learning models (Random Forest, SVM, and LSTM) to predict student risk levels.

The experimental results on a dataset of 423 students over one academic semester demonstrate that the LSTM-based model achieves an overall accuracy of 87.3%, a High Risk class recall of 91.2%, and an AUC-ROC of 0.93 for the High Risk class. These results represent a significant improvement over prior work and establish the feasibility of automated, non-intrusive burnout prediction using passively collected behavioural data.

The key contributions of this project are:

- A comprehensive multi-modal feature engineering framework that combines LMS engagement metrics, temporal behavioural patterns, submission behaviour indicators, and micro-survey responses into a unified 24-feature representation.
- An LSTM-based sequence model that effectively captures temporal dependencies in student behavioural data, outperforming traditional static classifiers on all evaluation metrics.
- An empirical analysis demonstrating that a 14-day observation window is sufficient for reliable burnout prediction, enabling early-semester risk identification.
- A modular, privacy-preserving system architecture designed for practical deployment within existing LMS infrastructure.

5.2 Limitations

The current system has several limitations that should be acknowledged. First, the dataset was collected from a single institution and may not generalise to students at different types of universities or in different cultural contexts. Second, the ground truth labels are derived from end-of-period surveys and counsellor assessments, which may not perfectly capture burnout states at earlier time points. Third, the system currently generates predictions at the day-week level and does not support real-time in-session monitoring.

5.3 Future Prospects

Several directions for future work are identified. First, the system should be validated on multi-institutional datasets to assess its generalisability. Second, the integration of additional data sources such as physiological signals from wearables and natural language features from student forum posts could further improve prediction accuracy. Third, the development of personalised prediction models that adapt to individual student baselines and trajectories is a promising research direction. Fourth, the design of evidence-based intervention strategies triggered by the system's predictions, in collaboration with academic counsellors, is a critical next step toward real-world impact. Finally, longitudinal evaluation of the system's effectiveness in reducing burnout prevalence in deployed settings remains an important open challenge.

5.4 Recommendations for Institutional Deployment

Based on the findings of this project, the following recommendations are made for institutions considering deploying the proposed system or similar automated burnout prediction frameworks:

- **Pilot before scaling:** Begin deployment in a single course or department with a small cohort of students and a dedicated academic advisor partner. Use the pilot to validate system performance in the local context, identify any data quality issues in the institution's LMS logs, and gather feedback from both students and advisors before scaling to the full institution.
- **Establish clear data governance policies:** Before deployment, institutions should establish and communicate clear policies governing what data is collected, how it is stored, who has access, how long it is retained, and under what circumstances it is deleted. These policies should be developed in consultation with data protection officers, student representatives, and academic staff.

- **Prioritise explainability in advisor interfaces:** Advisors should not be expected to act on unexplained risk scores. The dashboard interface should always display the top contributing factors to each student's risk classification, allowing advisors to corroborate the system's assessment with their own observations and to tailor their intervention accordingly.
- **Invest in counsellor capacity alongside the system:** An automated prediction system can only deliver benefit if the institution has sufficient counselling and support capacity to respond to the alerts it generates. Deploying a prediction system without expanding support capacity risks creating a situation where students are identified as at-risk but are placed on long waiting lists for counselling, which may worsen their distress.
- **Conduct regular model retraining and bias auditing:** The behavioural patterns associated with burnout may shift over time as academic culture, LMS platforms, and student demographics evolve. Models should be retrained periodically on fresh data, and each new model version should be subjected to a bias audit examining performance across demographic subgroups before deployment.
- **Obtain informed consent from students:** While LMS interaction data is routinely collected by institutions for administrative purposes, using it for mental health prediction represents a new use case that students may not anticipate. Institutions should obtain explicit informed consent from students before enrolling them in the prediction system, and should provide clear opt-out mechanisms.

5.5 Broader Implications

The framework developed in this project has implications that extend beyond academic burnout prediction. The core methodology, combining multi-modal behavioural data collection, temporal feature engineering, and LSTM-based sequence modelling, is applicable to a broad range of student well-being monitoring tasks including detection of social isolation, academic dishonesty risk, and mental health deterioration more generally.

More broadly, the project contributes to the emerging field of Positive Computing, which seeks to leverage technology not merely to increase productivity or efficiency but to actively promote human well-being. By creating systems that detect and respond to student distress, institutions can use technology to fulfil their duty of care toward their students in a scalable and evidence-based manner.

The project also has relevance beyond educational settings. The same behavioural analysis techniques can be applied to professional environments where digital footprints are generated continuously, such as remote workplaces. Employee burnout is estimated to cost the global economy over USD 300 billion annually in lost productivity, absenteeism, and healthcare costs [36]. Systems capable of early detection and intervention in professional settings could generate substantial social and economic benefits, making this a fertile area for future research and commercialisation.

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